SPACE DEBRIS TRACKING SIMULATOR

Simulation and Tracking of Low Earth Orbit (LEO) Debris with a Conceptual Magnetic Capture Mechanism

Semester 6 Project Report

B.Tech Aerospace Engineering

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1. Introduction

The growing population of artificial objects in Low Earth Orbit (LEO) presents an increasing collision hazard to operational satellites and future space missions. This project presents a MATLAB-based Space Debris Tracking Simulator that models the orbital motion of tracked debris fragments, evaluates their collision risk, classifies their material composition, simulates a conceptual magnetic capture mechanism for metallic debris, and models post-capture atmospheric-drag deorbit behaviour.

The simulator was developed as a Semester 6 project to gain hands-on exposure to space systems simulation, orbital mechanics, and MATLAB-based numerical modeling, under the guidance of Dr. Prashant Kumar.

2. Objectives

- Simulate orbital motion and state prediction of multiple LEO debris objects.
- Assess relative velocity and screen for potential collision (conjunction) events.
- Model a magnetic field-based capture concept for metallic debris fragments.
- Classify debris material (metallic vs. non-metallic) from simulated sensor features.
- Model atmospheric-drag deorbit and disposal behaviour of captured/decaying debris.

3. Methodology

The simulator is organized into five interconnected modules, described below.

3.1 Orbital Motion and State Prediction

Each debris object is initialized with a set of classical (Keplerian) orbital elements — semi-major axis (a), eccentricity (e), inclination (i), right ascension of the ascending node (Ω), argument of perigee (ω), and true anomaly (ν) — sampled for representative LEO altitudes between 300 km and 1500 km. These elements are converted to Earth-Centred Inertial (ECI) position and velocity vectors using the perifocal-to-ECI rotation:

$$r_{e}{}^{c}{}_{i} = Q_1 Q_2 Q_3 \cdot r_{p\ x}, \qquad v_{e}{}^{c}{}_{i} = Q_1 Q_2 Q_3 \cdot v_{p\ x}$$

where Q_1, Q_2, and Q_3 are the rotation matrices for Ω, i, and ω respectively. The two-body equations of motion are then numerically integrated (MATLAB ode45, RelTol = AbsTol = 1e-9):

$$d^2r/dt^2 = -(\mu \,/\, |r|^3) \cdot r$$

where μ = 398600.4418 km^3/s^2 is Earth's gravitational parameter. The instantaneous semi-major axis at each time step is recovered from the vis-viva equation:

$$1/a = 2/r - v^2/\mu$$

3.2 Velocity and Collision Assessment

For every pair of debris objects, the relative position and relative velocity vectors are computed at each time step:

$$\Delta r(t) = r_i(t) - r_j(t), \quad \Delta v(t) = v_i(t) - v_j(t)$$

The minimum separation distance over the propagated window is identified as the closest approach. If this miss distance falls below a screening threshold (5 km in this simulation), the pair is flagged as a conjunction event, and the corresponding relative speed and time of closest approach are logged.

3.3 Magnetic Field-Based Debris Capture Simulation

Metallic debris fragments are assumed to be captured through a conceptual on-axis magnetic dipole interaction with a chaser platform. The magnetic flux density at a separation distance d from a dipole of moment m is modeled as:

$$B(d) = (\mu_0 / 4\pi) \cdot (2m / d^3)$$

where $\mu_0 = 4\pi\times10^{-7}$ T·m/A is the permeability of free space. A debris object is considered captured once it enters a defined capture envelope radius and the local field strength exceeds a trigger threshold ($B > 1\times10^{-6}$ T in this simulation).

3.4 Material Classification of Debris

Each tracked object is assigned two synthetic sensor-derived features: radar cross-section (RCS, m²) and a magnetic susceptibility index (0–1). A nearest-centroid classifier assigns each object to a Metallic or Non-metallic class by comparing its feature vector to the mean feature vector (centroid) of each class:

$$class(k) = argmin^c \| [RCS_k, susc_k] - centroid^c \|$$

3.5 Deorbit and Disposal Modeling

Post-capture (or natural) orbital decay is modeled using an exponential atmospheric density profile and an averaged secular drag-decay approximation of King–Hele form:

$$\rho(h) = \rho_0 \cdot exp[-(h - h_0)/H], \quad da/dt = -C\_d\ (A/m)\ \rho\ v\ a$$

where C_d is the drag coefficient (2.2), A/m is the area-to-mass ratio (0.01 m²/kg), ρ_0 is the reference density at 400 km altitude (5.25×10^{-13} kg/m³), H is the atmospheric scale height (60 km), and v is the circular orbital speed. Decay is propagated in daily steps until the object reaches the re-entry interface altitude of 120 km.

4. Simulation Parameters and Data Used

Table 1 summarizes the key constants and simulation parameters used across all modules.

Table 1: Simulation constants and parameters

Parameter	Symbol	Value	Unit
Earth gravitational parameter	μ	398600.4418	km³/s²
Earth radius	R_E	6378.137	km
Number of tracked debris	N	8	—

Parameter	Symbol	Value	Unit
Debris altitude range	h	300 – 1500	km
Simulation duration	t	6000	s
Output time step	Δt	10	s
Collision screening threshold	—	5	km
Chaser magnetic dipole moment	m	500	$A \cdot m^2$
Capture envelope radius	—	0.05	km
Capture field trigger threshold	B	1×10^{-6}	T
Drag coefficient	C_d	2.2	—
Area-to-mass ratio	A/m	0.01	m^2/kg
Reference density (400 km)	ρ_0	5.25×10^{-13}	kg/m^3
Atmospheric scale height	H	60	km
Re-entry interface altitude	—	120	km

5. Results and Discussion

The debris field was propagated for a single simulation window and post-processed to generate the orbital, collision-screening, classification, and decay results summarized below.

5.1 Orbital Motion and Debris Field

Figure 1 shows the initial positions (red markers) of the eight tracked debris objects together with their propagated ground tracks over the simulation window, plotted against Earth (shown to scale).

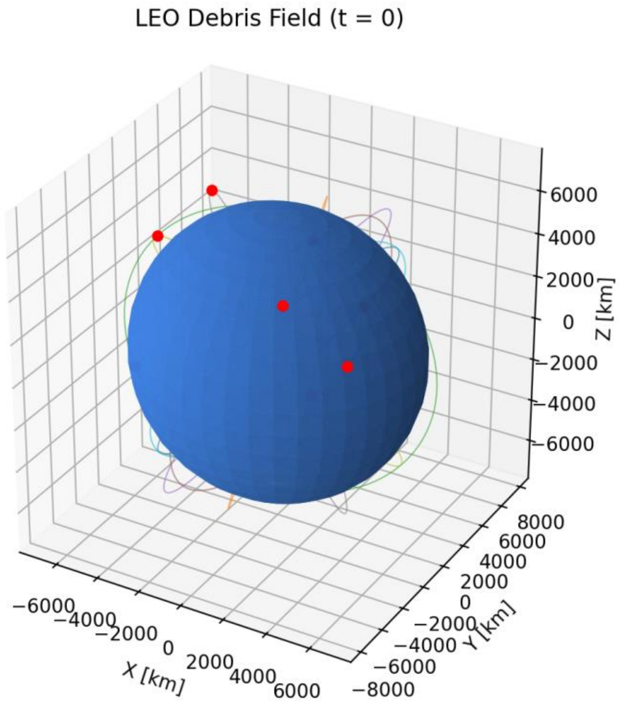

Figure 1: LEO debris field – orbital snapshot with propagated ground tracks

5.2 Altitude and Semi-Major Axis History

Figures 2 and 3 show the altitude and instantaneous semi-major axis of each debris object over the propagation window. As expected for unperturbed two-body motion, the semi-major axis of each object remains essentially constant, while altitude oscillates according to each object's eccentricity.

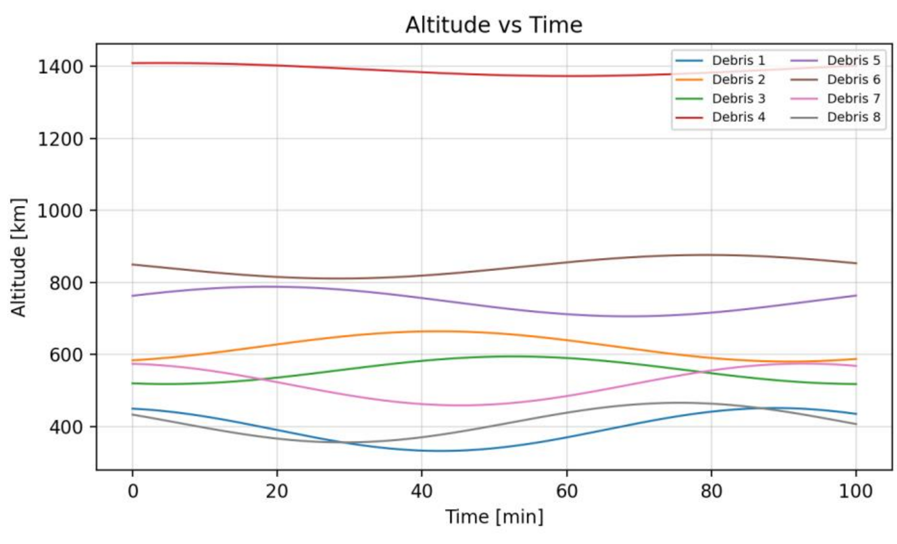

Figure 2: Altitude vs time for all tracked debris objects

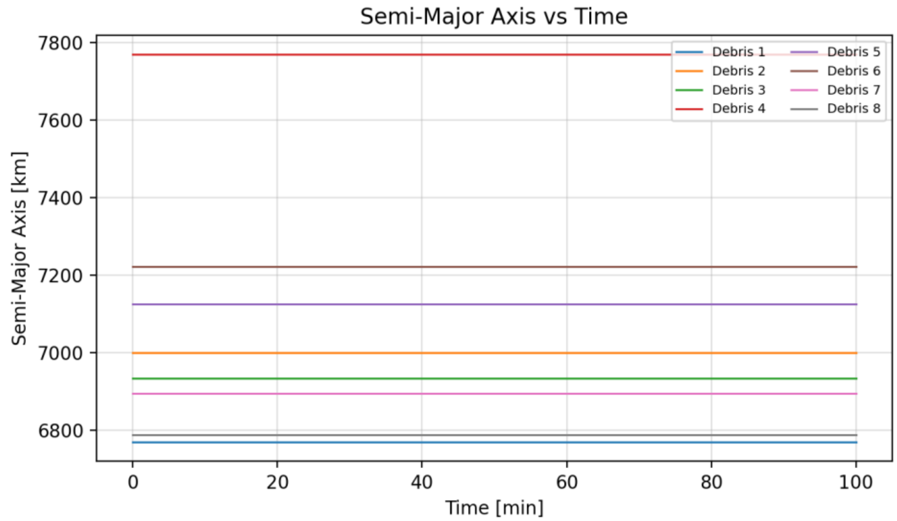

Figure 3: Semi-major axis vs time for all tracked debris objects

5.3 Collision Screening

Pairwise closest-approach screening was performed across all debris combinations. Any pair whose minimum separation fell below the 5 km screening threshold was flagged with its miss distance, relative speed, and time of closest approach, allowing conjunction risk to be prioritized.

5.4 Material Classification

Each debris object's simulated radar cross-section and magnetic susceptibility were passed through the nearest-centroid classifier described in Section 3.4. Figure 4 shows the resulting classification, separating metallic fragments (candidates for magnetic capture) from non-metallic fragments.

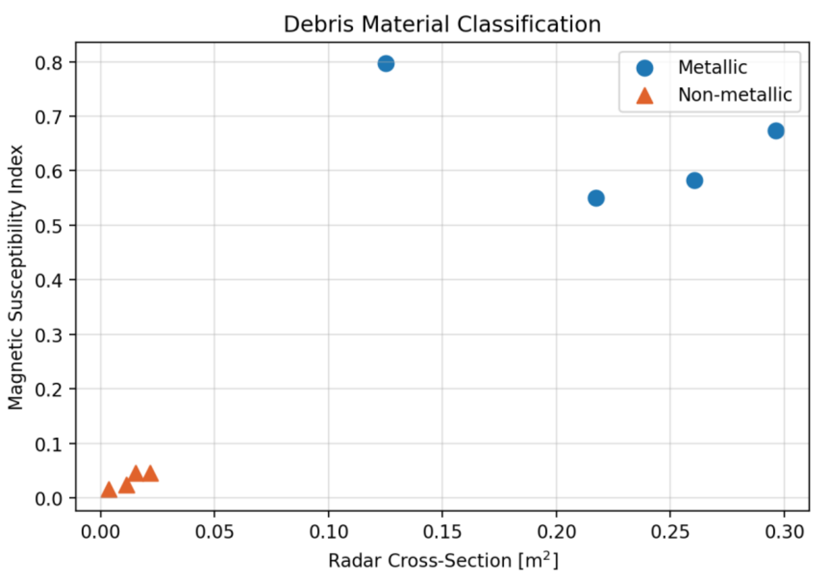

Figure 4: Debris material classification (RCS vs magnetic susceptibility)

5.5 Deorbit Decay

Figure 5 shows the modeled altitude decay of a representative object starting at 450 km altitude under atmospheric drag, illustrating the timescale (days) over which uncontrolled LEO objects in this altitude band naturally decay toward the 120 km re-entry interface.

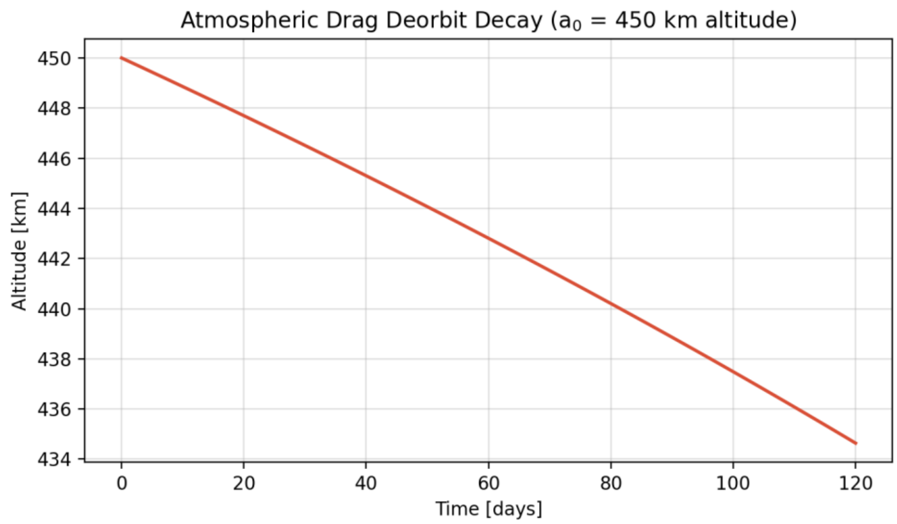

Figure 5: Atmospheric drag deorbit decay curve (a_0 = 450 km altitude)

6. Conclusion

This project demonstrated an integrated simulation pipeline for LEO debris tracking, combining two-body orbital propagation, collision-risk screening, a conceptual magnetic capture mechanism, sensor-based material classification, and atmospheric-drag deorbit modeling. The exercise provided valuable exposure to space systems simulation, orbital mechanics, and MATLAB-based numerical modeling techniques, and highlighted the practical challenges of active debris removal concepts such as magnetic capture for metallic fragments.

7. Acknowledgements

The authors gratefully acknowledge the continuous guidance and support of Dr. Prashant Kumar throughout the course of this project.

8. References

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